

Recall: Inclusion–Exclusion & Bonferroni

Given sets $A_1, \dots, A_n$ in a universe S , the **inclusion–exclusion principle** states

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \sum_{\substack{I \subseteq \{1, \dots, n\} \\ |I|=k}} \left| \bigcap_{i \in I} A_i \right|.$$

The **Bonferroni inequalities** say: truncating the right-hand side at the first q terms gives

- an *upper bound* on $|\bigcup A_i|$ when q is even, and
- a *lower bound* on $|\bigcup A_i|$ when q is odd.

Practice: Textbook Exercise 3.8.2 — Prove the Bonferroni inequalities

Prove the Bonferroni inequalities. If you cannot handle the general case, try at least the cases $q = 1$ and $q = 2$.

Remarks

Strategy. For each element x in the universe, let $c(x) = \#\{i : x \in A_i\}$. The contribution of x to the truncated sum at level q is $S_q(c(x)) = \sum_{k=1}^q (-1)^{k-1} \binom{c(x)}{k}$. The problem reduces to showing: for every integer $c \geq 1$,

$$S_q(c) \geq 1 \text{ if } q \text{ is odd,} \quad S_q(c) \leq 1 \text{ if } q \text{ is even.}$$

A useful identity: $\sum_{k=0}^q (-1)^k \binom{c}{k} = (-1)^q \binom{c-1}{q}$ for $c \geq 1$. (Prove it by induction on q using Pascal's rule.)

 Practice: Textbook Exercise 3.8.5 — Forbidden subsequences in permutations

How many orderings (permutations) of the 16 letters

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P

are there such that we *cannot* obtain any of the words **BAD**, **DEAF**, **APE** by crossing out some letters (i.e. as a *subsequence*)?

What if we also forbid **LEADING**?



Remarks

Key observation. A word $w_1w_2\cdots w_k$ appears as a subsequence of a permutation iff the letters $w_1, w_2, \dots, w_k$ appear in that relative order. Among all $k!$ orderings of those letters, exactly one gives the forbidden pattern. For intersections, identify the combined partial-order constraints on the involved letters and count *linear extensions* of the resulting poset. Watch out for pairs of constraints that *contradict* each other (e.g. one requires A before D , another requires D before A).